Lecture 5: Proof Notes for Board Work

ISE 5406 – Optimization Theory

These notes contain the proofs to write on the board during lecture.
Green = Steps **Orange** = Key terms **Purple** = Results/Conclusions

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1. Closest-Point Theorem

Statement

Let $S \subseteq \mathbb{R}^n$ be nonempty, **closed**, and **convex**. For any $\mathbf{y} \notin S$, there exists a **unique** point $\bar{\mathbf{x}} \in S$ closest to $\mathbf{y}$. Moreover, $\bar{\mathbf{x}}$ is closest iff $(\mathbf{y} - \bar{\mathbf{x}})^T(\mathbf{x} - \bar{\mathbf{x}}) \leq 0$ for all $\mathbf{x} \in S$.

Step 1: Existence

Define $f(\mathbf{x}) = \|\mathbf{y} - \mathbf{x}\|^2$ for $\mathbf{x} \in S$

This is a continuous function on S.

[Need to show a minimizer exists]

Take any $\mathbf{x}_0 \in S$. The sublevel set $\{\mathbf{x} \in S : f(\mathbf{x}) \leq f(\mathbf{x}_0)\}$ is:

- **Closed**: intersection of closed sets
- **Bounded**: contained in ball of radius $\|\mathbf{y} - \mathbf{x}_0\|$ centered at $\mathbf{y}$

$\Rightarrow$ Sublevel set is **compact**

By Weierstrass theorem: continuous function on compact set attains minimum

$\boxed{\exists \bar{\mathbf{x}} \in S \text{ minimizing } \|\mathbf{y} - \mathbf{x}\| \checkmark}$

Step 2: Uniqueness

Suppose $\bar{\mathbf{x}}_1, \bar{\mathbf{x}}_2 \in S$ both minimize distance to $\mathbf{y}$

Let $d = \|\mathbf{y} - \bar{\mathbf{x}}_1\| = \|\mathbf{y} - \bar{\mathbf{x}}_2\|$

Consider $\bar{\mathbf{x}}_3 = \frac{1}{2}(\bar{\mathbf{x}}_1 + \bar{\mathbf{x}}_2)$

By convexity of S: $\bar{\mathbf{x}}_3 \in S$

[Now show $\bar{\mathbf{x}}_3$ is strictly closer to $\mathbf{y}$ unless $\bar{\mathbf{x}}_1 = \bar{\mathbf{x}}_2$]

$\|\mathbf{y} - \bar{\mathbf{x}}_3\|^2 = \left\|\mathbf{y} - \frac{\bar{\mathbf{x}}_1 + \bar{\mathbf{x}}_2}{2}\right\|^2 = \left\|\frac{(\mathbf{y} - \bar{\mathbf{x}}_1) + (\mathbf{y} - \bar{\mathbf{x}}_2)}{2}\right\|^2$

Using $\|a+b\|^2 = \|a\|^2 + 2a^T b + \|b\|^2$:

$= \frac{1}{4}\left(\|\mathbf{y}-\bar{\mathbf{x}}_1\|^2 + 2(\mathbf{y}-\bar{\mathbf{x}}_1)^T(\mathbf{y}-\bar{\mathbf{x}}_2) + \|\mathbf{y}-\bar{\mathbf{x}}_2\|^2\right)$

$= \frac{d^2}{2} + \frac{1}{2}(\mathbf{y}-\bar{\mathbf{x}}_1)^T(\mathbf{y}-\bar{\mathbf{x}}_2)$

By Cauchy-Schwarz: $(\mathbf{y}-\bar{\mathbf{x}}_1)^T(\mathbf{y}-\bar{\mathbf{x}}_2) \le d^2$ with equality iff $\mathbf{y}-\bar{\mathbf{x}}_1 = \mathbf{y}-\bar{\mathbf{x}}_2$

$\Rightarrow \quad \|\mathbf{y}-\bar{\mathbf{x}}_3\|^2 \le d^2$ with equality iff $\bar{\mathbf{x}}_1 = \bar{\mathbf{x}}_2$

Unique closest point ✓

Step 3: Characterization ($\Rightarrow$)

Assume $\bar{\mathbf{x}}$ minimizes $\|\mathbf{y}-\mathbf{x}\|$ over S

Let $\mathbf{x} \in S$ and $\lambda \in (0,1]$. By convexity: $(1-\lambda)\bar{\mathbf{x}} + \lambda\mathbf{x} \in S$

$\|\mathbf{y}-\bar{\mathbf{x}}\|^2 \le \|\mathbf{y}-((1-\lambda)\bar{\mathbf{x}}+\lambda\mathbf{x})\|^2$

$= \|\mathbf{y}-\bar{\mathbf{x}}-\lambda(\mathbf{x}-\bar{\mathbf{x}})\|^2$

$= \|\mathbf{y}-\bar{\mathbf{x}}\|^2 - 2\lambda(\mathbf{y}-\bar{\mathbf{x}})^T(\mathbf{x}-\bar{\mathbf{x}}) + \lambda^2\|\mathbf{x}-\bar{\mathbf{x}}\|^2$

$\Rightarrow \quad 0 \le -2\lambda(\mathbf{y}-\bar{\mathbf{x}})^T(\mathbf{x}-\bar{\mathbf{x}}) + \lambda^2\|\mathbf{x}-\bar{\mathbf{x}}\|^2$

Divide by $\lambda > 0$ and let $\lambda \to 0$:

$(\mathbf{y}-\bar{\mathbf{x}})^T(\mathbf{x}-\bar{\mathbf{x}}) \le 0$

Characterization ($\Rightarrow$) proved ✓

Step 4: Characterization ($\Leftarrow$)

Assume $(\mathbf{y}-\bar{\mathbf{x}})^T(\mathbf{x}-\bar{\mathbf{x}}) \le 0$ for all $\mathbf{x} \in S$

For any $\mathbf{x} \in S$:

$\|\mathbf{y}-\mathbf{x}\|^2 = \|(\mathbf{y}-\bar{\mathbf{x}}) + (\bar{\mathbf{x}}-\mathbf{x})\|^2$

$= \|\mathbf{y}-\bar{\mathbf{x}}\|^2 + 2(\mathbf{y}-\bar{\mathbf{x}})^T(\bar{\mathbf{x}}-\mathbf{x}) + \|\bar{\mathbf{x}}-\mathbf{x}\|^2$

$= \|\mathbf{y}-\bar{\mathbf{x}}\|^2 - 2(\mathbf{y}-\bar{\mathbf{x}})^T(\mathbf{x}-\bar{\mathbf{x}}) + \|\bar{\mathbf{x}}-\mathbf{x}\|^2$

Since $(\mathbf{y}-\bar{\mathbf{x}})^T(\mathbf{x}-\bar{\mathbf{x}}) \le 0$: the middle term is ≥ 0

$\Rightarrow \quad \|\mathbf{y}-\mathbf{x}\|^2 \ge \|\mathbf{y}-\bar{\mathbf{x}}\|^2$

$\bar{\mathbf{x}}$ is closest point □

2. Separating Hyperplane Theorem

Statement

Let $S \subseteq \mathbb{R}^n$ be nonempty, closed, and **convex**. For any $\mathbf{y} \notin S$, there exists $\mathbf{p} \neq \mathbf{0}$ and α such that $\mathbf{p}^T\mathbf{y} > \alpha$ and $\mathbf{p}^T\mathbf{x} \leq \alpha$ for all $\mathbf{x} \in S$.

Step 1: Find the closest point

By Closest-Point Theorem: $\exists$ unique $\bar{\mathbf{x}} \in S$ closest to $\mathbf{y}$

Step 2: Define the separating hyperplane

Let $\mathbf{p} = \mathbf{y} - \bar{\mathbf{x}}$

Note: $\mathbf{p} \neq \mathbf{0}$ since $\mathbf{y} \notin S$ but $\bar{\mathbf{x}} \in S$

Let $\alpha = \mathbf{p}^T\bar{\mathbf{x}}$

Step 3: Verify $\mathbf{p}^T\mathbf{y} > \alpha$

$\mathbf{p}^T\mathbf{y} = (\mathbf{y} - \bar{\mathbf{x}})^T\mathbf{y} = \mathbf{y}^T\mathbf{y} - \bar{\mathbf{x}}^T\mathbf{y}$

$\alpha = \mathbf{p}^T\bar{\mathbf{x}} = (\mathbf{y} - \bar{\mathbf{x}})^T\bar{\mathbf{x}} = \mathbf{y}^T\bar{\mathbf{x}} - \bar{\mathbf{x}}^T\bar{\mathbf{x}}$

$\mathbf{p}^T\mathbf{y} - \alpha = \|\mathbf{y}\|^2 - \bar{\mathbf{x}}^T\mathbf{y} - \mathbf{y}^T\bar{\mathbf{x}} + \|\bar{\mathbf{x}}\|^2 = \|\mathbf{y} - \bar{\mathbf{x}}\|^2 > 0$

[Positive since $\mathbf{y} \neq \bar{\mathbf{x}}$]

$\boxed{\mathbf{p}^T\mathbf{y} > \alpha \ \checkmark}$

Step 4: Verify $\mathbf{p}^T\mathbf{x} \leq \alpha$ for all $\mathbf{x} \in S$

By characterization from Closest-Point Theorem:

$(\mathbf{y} - \bar{\mathbf{x}})^T(\mathbf{x} - \bar{\mathbf{x}}) \leq 0$ for all $\mathbf{x} \in S$

$\mathbf{p}^T(\mathbf{x} - \bar{\mathbf{x}}) \leq 0$

$\Rightarrow \quad \mathbf{p}^T\mathbf{x} \leq \mathbf{p}^T\bar{\mathbf{x}} = \alpha$

$\boxed{\mathbf{p}^T\mathbf{x} \leq \alpha \text{ for all } \mathbf{x} \in S \ \square}$

3. Supporting Hyperplane Theorem

Statement

Let $S \subseteq \mathbb{R}^n$ be nonempty and **convex**. For any $\bar{\mathbf{x}} \in \partial S$ (boundary of S), there exists a **supporting hyperplane** at $\bar{\mathbf{x}}$, i.e., $\mathbf{p} \neq \mathbf{0}$ with $\mathbf{p}^T(\mathbf{x} - \bar{\mathbf{x}}) \leq 0$ for all $\mathbf{x} \in S$.

Step 1: Construct sequence approaching boundary point

Since $\bar{\mathbf{x}} \in \partial S$: every neighborhood of $\bar{\mathbf{x}}$ contains points outside S

Choose $\mathbf{y}_k \notin S$ with $\mathbf{y}_k \to \bar{\mathbf{x}}$ as $k \to \infty$

[e.g., $\mathbf{y}_k \in B(\bar{\mathbf{x}}, 1/k) \setminus S$]

Step 2: Apply Separating Hyperplane Theorem to each $\mathbf{y}_k$

For each k: $\mathbf{y}_k \notin \mathrm{cl}(S)$ *[or if S is closed, $\mathbf{y}_k \notin S$]*

By Separation Theorem: $\exists \mathbf{p}_k$ with $\mathbf{p}_k^T \mathbf{y}_k > \mathbf{p}_k^T \mathbf{x}$ for all $\mathbf{x} \in S$

Normalize: $\|\mathbf{p}_k\| = 1$

Step 3: Extract convergent subsequence

$\{\mathbf{p}_k\}$ is a sequence on the unit sphere (compact)

By Bolzano-Weierstrass: $\exists$ convergent subsequence $\mathbf{p}_{k_j} \to \mathbf{p}$

Note: $\|\mathbf{p}\| = 1$, so $\mathbf{p} \neq \mathbf{0}$

Step 4: Pass to limit

From $\mathbf{p}_{k_j}^T \mathbf{y}_{k_j} \geq \mathbf{p}_{k_j}^T \mathbf{x}$ for all $\mathbf{x} \in S$

Taking $j \to \infty$:

$\mathbf{p}^T \bar{\mathbf{x}} \geq \mathbf{p}^T \mathbf{x}$ for all $\mathbf{x} \in S$

Rearranging: $\mathbf{p}^T(\mathbf{x} - \bar{\mathbf{x}}) \leq 0$ for all $\mathbf{x} \in S$

$H = \{\mathbf{x} : \mathbf{p}^T(\mathbf{x} - \bar{\mathbf{x}}) = 0\}$ **is a supporting hyperplane at** $\bar{\mathbf{x}}$ $\square$

4. Local Implies Global Theorem

Statement

Let $f : S \to \mathbb{R}$ be **convex** on **convex** set S. If $\bar{\mathbf{x}}$ is a **local minimum**, then $\bar{\mathbf{x}}$ is a **global minimum**.

Step 1: State the setup

$\bar{\mathbf{x}}$ is local min: $\exists \epsilon > 0$ s.t. $f(\bar{\mathbf{x}}) \leq f(\mathbf{x})$ for all $\mathbf{x} \in S \cap B(\bar{\mathbf{x}}, \epsilon)$

Goal: Show $f(\bar{\mathbf{x}}) \leq f(\mathbf{x})$ for all $\mathbf{x} \in S$

Step 2: Handle points close to $\bar{\mathbf{x}}$

For $\mathbf{x} \in S$ with $\|\mathbf{x} - \bar{\mathbf{x}}\| < \epsilon$:

$f(\bar{\mathbf{x}}) \leq f(\mathbf{x})$ by local optimality ✓

Step 3: Handle points far from $\bar{\mathbf{x}}$

For $\mathbf{x} \in S$ with $\|\mathbf{x} - \bar{\mathbf{x}}\| \geq \epsilon$:

Choose $\lambda = \frac{\epsilon}{2\|\mathbf{x} - \bar{\mathbf{x}}\|} \in (0, 1)$

Define $\mathbf{y} = (1 - \lambda)\bar{\mathbf{x}} + \lambda \mathbf{x}$

[$\mathbf{y}$ is on line segment from $\bar{\mathbf{x}}$ to $\mathbf{x}$, inside the ϵ-ball]

Step 4: Verify $\mathbf{y}$ is in the local neighborhood

$\|\mathbf{y} - \bar{\mathbf{x}}\| = \|\lambda(\mathbf{x} - \bar{\mathbf{x}})\| = \lambda\|\mathbf{x} - \bar{\mathbf{x}}\| = \frac{\epsilon}{2} < \epsilon$

$\Rightarrow \quad \mathbf{y} \in B(\bar{\mathbf{x}}, \epsilon) \cap S$

So $f(\bar{\mathbf{x}}) \leq f(\mathbf{y})$ by local optimality

Step 5: Apply convexity

By **convexity** of f:

$f(\mathbf{y}) = f((1 - \lambda)\bar{\mathbf{x}} + \lambda \mathbf{x}) \leq (1 - \lambda) f(\bar{\mathbf{x}}) + \lambda f(\mathbf{x})$

Step 6: Combine inequalities

From Steps 4-5:

$f(\bar{\mathbf{x}}) \leq f(\mathbf{y}) \leq (1 - \lambda) f(\bar{\mathbf{x}}) + \lambda f(\mathbf{x})$

$f(\bar{\mathbf{x}}) \leq (1 - \lambda) f(\bar{\mathbf{x}}) + \lambda f(\mathbf{x})$

Rearranging: $f(\bar{\mathbf{x}}) - (1 - \lambda) f(\bar{\mathbf{x}}) \leq \lambda f(\mathbf{x})$

$\lambda f(\bar{\mathbf{x}}) \leq \lambda f(\mathbf{x})$

Since $\lambda > 0$: divide by λ

$\boxed{f(\bar{\mathbf{x}}) \leq f(\mathbf{x}) \textbf{ for all } \mathbf{x} \in S \ \square}$

5. Optimality Conditions Theorem

Statement

Let $f : S \to \mathbb{R}$ be **convex** on **convex** set S. Then $\bar{\mathbf{x}} \in S$ is optimal iff $\exists$ **subgradient** $\boldsymbol{\xi}$ at $\bar{\mathbf{x}}$ with $\boldsymbol{\xi}^T(\mathbf{x} - \bar{\mathbf{x}}) \geq 0$ for all $\mathbf{x} \in S$.

Step 1: ($\Leftarrow$) Sufficient condition

Assume $\exists\boldsymbol{\xi}$ with $\boldsymbol{\xi}^T(\mathbf{x} - \bar{\mathbf{x}}) \geq 0$ for all $\mathbf{x} \in S$

By **subgradient definition**:

$f(\mathbf{x}) \geq f(\bar{\mathbf{x}}) + \boldsymbol{\xi}^T(\mathbf{x} - \bar{\mathbf{x}})$ for all $\mathbf{x} \in S$

Since $\boldsymbol{\xi}^T(\mathbf{x} - \bar{\mathbf{x}}) \geq 0$:

$f(\mathbf{x}) \geq f(\bar{\mathbf{x}}) + 0 = f(\bar{\mathbf{x}})$

$\boxed{\bar{\mathbf{x}} \text{ is optimal } \checkmark}$

Step 2: ($\Rightarrow$) Necessary condition – Interior case

Assume $\bar{\mathbf{x}}$ is optimal and $\bar{\mathbf{x}} \in \text{int}(S)$

Since $\bar{\mathbf{x}}$ is interior: $\exists\boldsymbol{\xi} \in \partial f(\bar{\mathbf{x}})$ (subgradient exists)

By subgradient inequality: $f(\mathbf{x}) \geq f(\bar{\mathbf{x}}) + \boldsymbol{\xi}^T(\mathbf{x} - \bar{\mathbf{x}})$ for all $\mathbf{x}$

Since $\bar{\mathbf{x}}$ is optimal: $f(\mathbf{x}) \geq f(\bar{\mathbf{x}})$ for all $\mathbf{x} \in S$

[At interior optimal point, the zero vector must be a subgradient]

For small $\epsilon > 0$: $\bar{\mathbf{x}} + \epsilon\mathbf{d} \in S$ for any direction $\mathbf{d}$

$f(\bar{\mathbf{x}} + \epsilon\mathbf{d}) \geq f(\bar{\mathbf{x}}) + \epsilon\boldsymbol{\xi}^T\mathbf{d}$

Since $\bar{\mathbf{x}}$ is optimal: $f(\bar{\mathbf{x}} + \epsilon\mathbf{d}) \geq f(\bar{\mathbf{x}})$

Subtracting: $0 \geq \epsilon\boldsymbol{\xi}^T\mathbf{d}$

This holds for all $\mathbf{d}$, so taking $\mathbf{d}$ and $-\mathbf{d}$: $\boldsymbol{\xi}^T\mathbf{d} = 0$

$\Rightarrow \quad \boldsymbol{\xi} = \mathbf{0}$, and $\mathbf{0}^T(\mathbf{x} - \bar{\mathbf{x}}) = 0 \geq 0$ for all $\mathbf{x}$

$\boxed{\textbf{Interior case: } \boldsymbol{\xi} = \mathbf{0} \textbf{ works } \checkmark}$

Step 3: ($\Rightarrow$) Necessary condition – Boundary case

Assume $\bar{\mathbf{x}}$ is optimal and $\bar{\mathbf{x}} \in \partial S$

Consider $\text{epi}(f) = \{(\mathbf{x}, z) : \mathbf{x} \in S, z \geq f(\mathbf{x})\}$

Point $(\bar{\mathbf{x}}, f(\bar{\mathbf{x}})) \in \partial(\text{epi}(f))$ *[boundary of epigraph]*

By **Supporting Hyperplane Theorem**:

$\exists(\boldsymbol{\xi}, -\mu) \neq \mathbf{0}$ with $(\boldsymbol{\xi}, -\mu)^T((\mathbf{x}, z) - (\bar{\mathbf{x}}, f(\bar{\mathbf{x}}))) \leq 0$

for all $(\mathbf{x}, z) \in \text{epi}(f)$

$\boldsymbol{\xi}^T(\mathbf{x} - \bar{\mathbf{x}}) - \mu(z - f(\bar{\mathbf{x}})) \leq 0$

Taking $\mathbf{x} = \bar{\mathbf{x}}$ and $z \to \infty$: need $\mu \geq 0$

If $\mu = 0$: $\boldsymbol{\xi}^T(\mathbf{x} - \bar{\mathbf{x}}) \leq 0$ for all $\mathbf{x} \in S$, take $\boldsymbol{\xi}' = -\boldsymbol{\xi}$

If $\mu > 0$: divide by μ and let $\boldsymbol{\xi}' = \boldsymbol{\xi}/\mu$

Taking $z = f(\mathbf{x})$: $\boldsymbol{\xi}'^T(\mathbf{x} - \bar{\mathbf{x}}) \leq f(\mathbf{x}) - f(\bar{\mathbf{x}})$

$\boldsymbol{\xi}'$ is subgradient with $\boldsymbol{\xi}'^T(\mathbf{x} - \bar{\mathbf{x}}) \geq 0$ (from optimality) $\square$

Summary: Key Theorems in Lecture 5

Proof Dependency Chain

Closest-Point Theorem
$\downarrow$
Separating Hyperplane Theorem
$\downarrow$
Supporting Hyperplane Theorem
$\downarrow$
Optimality Conditions

Key ideas to emphasize:

- **Closest-Point:** Uses compactness for existence, strict convexity of $\|\cdot\|^2$ for uniqueness, first-order optimality for characterization.
- **Separation:** Closest point gives normal direction; distance gives strict separation.
- **Supporting:** Limit argument using normalized separating normals; compactness of unit sphere.
- **Local Implies Global:** Convexity lets us "reach" any point via the local neighborhood using line segments.
- **Optimality Conditions:** Sufficient is easy (just use subgradient definition). Necessary uses supporting hyperplane of epigraph.

Time estimates:

- Closest-Point: 8-10 minutes (most detailed)
- Separation: 5-7 minutes (builds on Closest-Point)
- Supporting: 6-8 minutes (limit argument needs care)
- Local Implies Global: 4-5 minutes (short and elegant)
- Optimality: 6-8 minutes (two directions)